

MOSAIC: Calibrated Per-Cell Uncertainty for Unpaired Single-Cell Multi-Omics Integration in Disease Atlases

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Abstract

When a patient biopsy is profiled by both RNA sequencing and chromatin accessibility sequencing, the two measurements describe overlapping cell populations through different molecular lenses — but each individual cell is destroyed at measurement, so no cell appears in both datasets. Mapping one cell’s RNA measurement to the most likely matching cell in the accessibility dataset (“unpaired integration”) is now a routine step in building disease atlases, yet every existing method returns a single best-guess match with no indication of how trustworthy that match is. A wrong match propagates silently into every downstream analysis: joint cell-type discovery, gene-regulatory inference, and disease-biomarker identification. We introduce MOSAIC (Multi-Omic Stochastic Alignment via Information-theoretic Coupling), which solves this problem by framing the matching as a probabilistic transport plan and then computing a per-cell *uncertainty score* — a number between 0 and 1 indicating whether a cell was aligned confidently or ambiguously. The key technical discovery is that the naive candidate uncertainty score (the row entropy of the transport plan) is directionally *wrong* for dense cell clusters; our fix — marginalizing the transport plan over cell-type cluster labels before computing entropy — costs one matrix-vector product and produces a calibrated signal. On three public benchmarks (PBMC immune cells, mouse brain development, CITE-seq protein+RNA), MOSAIC outperforms all baselines on every alignment metric, detects wrong-cluster assignments with AUROC 0.88–0.95, and — most clinically relevant — flags cell populations that exist in a diseased sample but are absent from the healthy reference with mean AUROC 0.96–0.995. A cross-tissue negative control (immune blood cells aligned against mouse brain cells, sharing no cell types) yields $4.2\times$ higher mean uncertainty than valid within-dataset alignments, confirming the signal correctly reports high uncertainty when nothing truly matches.

1. Introduction

Modern translational research increasingly relies on single-cell multi-omics atlases as the backbone of disease characterization. The premise is simple but powerful: by measuring the molecular state of individual cells across thousands of patients — their gene expression (RNA), their chromatin accessibility (which DNA regions are open and potentially regulatory), and their surface protein levels — we can build a reference map of every cell type in every disease state, and then use that map to generate therapeutic hypotheses. The Human Cell Atlas ([Regev et al., 2017](#)), the Human Tumor Atlas Network ([Rozenblatt-Rosen et al., 2020](#)), the COVID-19 Cell Atlas ([Stephenson et al., 2021](#)), and the HuBMAP project ([HuBMAP Consortium, 2019](#)) all operate on this premise.

A central practical obstacle is that the relevant measurements are almost never taken on the same cell. Single-cell RNA sequencing (scRNA-seq) measures gene expression by lysing each cell and reading its messenger RNA; single-cell ATAC-seq (scATAC-seq) measures chromatin accessibility by fragmenting DNA in each cell; the cell is destroyed in both cases. Paired assays that take both measurements from one cell (such as 10x Multiome and CITE-seq) exist but remain expensive and cover only a small fraction of existing atlas data. The vast majority of disease atlases are therefore *unpaired*: the RNA dataset and the ATAC dataset describe overlapping cell populations, but no individual cell appears in both. Unpaired integration — computationally matching a cell measured in one modality to its most likely counterpart in another — has become a standard preprocessing step for joint cell-type discovery, gene-regulatory inference, cell-communication analysis, and disease-association studies (Demetci et al., 2022; Cao and Gao, 2022; Cao et al., 2022; Hao et al., 2024).

A clinically consequential gap pervades this literature: no existing method reports *per-cell uncertainty* about the match it produces. Think of unpaired integration like a match-making service that pairs RNA profiles to ATAC profiles — but never tells you how confident the match is, or whether a particular pairing is likely wrong. Existing approaches — Seurat v5 (Hao et al., 2024), GLUE (Cao and Gao, 2022), SCOT (Demetci et al., 2022), uniPort (Cao et al., 2022), MultiVI (Ashuach et al., 2023), LIGER (Welch et al., 2019), Harmony (Korsunsky et al., 2019) — all return a point estimate: their best guess, with no confidence attached. The downstream consequence is serious. Incorrect alignments propagate silently into joint cell-type labels, differential-accessibility tests, enhancer–gene linkage inferences, and spatial deconvolution — the steps that produce therapeutic hypotheses. A candidate regulatory program that is in fact an alignment artifact can redirect a pharmaceutical program by years. More importantly, a disease atlas routinely contains cell populations that do not exist in the healthy reference (consider a tumor-specific T-cell exhaustion state, or a neurodegeneration-specific reactive astrocyte subtype). Without uncertainty estimates, these novel disease cells are silently matched to the nearest available healthy population, producing a spurious null finding precisely where clinical novelty is most valuable.

We introduce MOSAIC (Multi-Omic Stochastic Alignment via Information-theoretic Coupling), a method that reframes unpaired single-cell multi-omics integration as a probabilistic problem and reports a cluster-resolved alignment entropy as a per-cell uncertainty measure. MOSAIC composes three components. Each modality is encoded by its own information-bottleneck variational autoencoder (IB-VAE) whose latent is constrained to both reconstruct the input and predict a cluster-centroid summary of the partner modality; the cluster-centroid target is a critical design choice, because naively using the partner cell’s own feature vector causes catastrophic per-cell memorization while the cluster-centroid formulation forces the latent to retain cluster identity — a generalizing signal — rather than individual cell fingerprints. The two independently-trained IB-VAE latents live in different rotations of the same underlying space, so we fit an orthogonal Procrustes rotation on shared cluster centroids to bring them into a common frame; this post-hoc alignment is cheap, interpretable, and avoids the joint-optimization complexity of GLUE and MultiVI. Finally we solve an entropic optimal-transport problem between the two latent clouds and, rather than reporting per-row entropy of the transport plan — which we show is miscalibrated as a per-cell uncertainty signal because of the geometry of dense clusters — we marginalize each

row over the partner modality’s cluster labels. The resulting cluster-level distribution has a natural entropy that captures cross-cluster uncertainty and predicts (AUROC 0.88–0.95) which cells have been assigned to the wrong cluster.

The core technical insight, phrased as a negative result, is that the obvious candidate uncertainty signal — the entropy (spread) of the probabilistic matching distribution for each cell — is not merely weak but directionally *wrong*. On PBMC immune-cell data it yields a Spearman correlation of -0.65 with true alignment error, meaning cells flagged as “high uncertainty” actually tend to be *correctly* matched. The reason is a geometric artifact: cells in dense clusters have many valid within-cluster match candidates (spreading out the distribution, inflating entropy) while simultaneously having a very close true partner (reducing actual error). Our fix is to compute entropy not over individual cells but over *cell-type cluster labels* — asking “does this cell’s transport mass concentrate on one cell type or spread across multiple?” rather than “does it concentrate on one specific cell?” This marginalization step costs one matrix-vector product and resolves the confound entirely. On three paired benchmarks — 10x Multiome PBMC 10k (11,303 immune cells, 18 cell types), 10x Multiome E18 mouse brain 5k (4,531 neural cells, 20 types), and 10x CITE-seq PBMC (RNA + 14 protein markers, 16 types) — MOSAIC outperforms all baselines on every alignment metric and identifies wrong-cluster cells with AUROC 0.88–0.95 (AUROC measures whether a ranking correctly separates two groups; 1.0 is perfect, 0.5 is chance). Most importantly for clinical use: in a leave-one-cluster-out experiment simulating a disease-specific cell type absent from the healthy reference, MOSAIC’s uncertainty score flags absent cells with mean AUROC 0.96 on PBMC, 0.995 on brain, and 0.946 on CITE-seq. A cross-tissue negative control (matching immune blood cells against mouse brain cells, which share zero cell types) shows $4.2\times$ higher mean cluster entropy than valid within-dataset alignments, confirming the signal correctly reports high uncertainty when nothing truly matches.

Generalizable Insights about Machine Learning in the Context of Healthcare

- **Most clinical disease atlases are unpaired — uncertainty quantification is therefore a first-class requirement, not an optional refinement.** Paired profiling assays are expensive and rare; the data powering real disease atlases is overwhelmingly single-modality per cell. A method that returns only a best-guess match leaves researchers unable to distinguish a trustworthy regulatory hypothesis from a silent artifact, which matters because a wrong match entering a drug-target pipeline can redirect a program by years.
- **Per-cell uncertainty unlocks three clinical use cases that alignment alone cannot provide:** (1) *filtering* — discard unreliable matches before regulatory inference or biomarker calling; (2) *triage* — flag high-confidence matches for downstream validation prioritization; (3) *disease-specific population detection* — identify cell types in the disease atlas that do not exist in the healthy reference (mean AUROC 0.96–0.995 in our leave-one-cluster-out experiments). None of these is possible with a point-estimate alignment.

- **The obvious uncertainty metric for probabilistic cell matching is directionally wrong.** Row entropy of an entropic optimal-transport plan anti-correlates with alignment error (Spearman $\rho = -0.65$ on PBMC) because dense clusters simultaneously inflate entropy and reduce true-partner distance. The fix — marginalizing over partner cluster labels before computing entropy — is one matrix-vector product and converts the signal from misleading to well-discriminating (AUROC 0.88–0.95). This pathology generalizes to any optimal-transport integration method that exposes per-row entropy as a confidence score.
- **When coupling two independently-encoded modalities, prediction targets must be cluster-level, not cell-level.** Using an individual cell’s partner-modality features as a training target causes catastrophic memorization (validation MSE $20\times$ training MSE); switching to the cluster-centroid target — the mean over all cells of the same type — forces the model to learn cluster identity rather than individual fingerprints, producing a $3.6\times$ improvement in cell-type label transfer. This principle applies to any multi-modal architecture that links two encoders through a cross-modal prediction head.

2. Related Work

The unpaired multi-omics integration literature is crowded and largely driven by biological-benchmarking criteria: SCOT (Demetci et al., 2022) uses Gromov–Wasserstein optimal transport on intra-modality distance matrices; uniPort (Cao et al., 2022) uses coupled autoencoders with a domain-adversarial regularizer; MultiVI (Ashuach et al., 2023) jointly trains a shared-latent VAE with modality-specific decoders; GLUE (Cao and Gao, 2022) uses a graph-linked unified embedding constrained by prior biological graphs (e.g., peak-to-gene linkages); LIGER (Welch et al., 2019) and Harmony (Korsunsky et al., 2019) use non-negative matrix factorization and iterative clustering; and Seurat v5 bridge integration (Hao et al., 2024) re-uses a small paired dataset as an anchoring bridge between two unpaired datasets. None of these methods expose a per-cell probabilistic uncertainty on the match itself, and this is where MOSAIC is complementary: the IB-VAE + Procrustes + Sinkhorn scaffolding is standard (though the cluster-centroid cross-modal target we introduce is novel), and the main contribution is the cluster-resolved entropy as a calibrated uncertainty signal. Adjacent literature on uncertainty quantification in single-cell analysis — scANVI (Xu et al., 2021) exposes posterior variances on latent embeddings, uncertainty quantification has been applied to atlas-level cell-type transfer (Engelmann et al., 2022) — does not translate into cell-to-cell match confidence across modalities; no prior method, to our knowledge, reports per-cell uncertainty on an unpaired multi-omics match. Optimal transport for single-cell data is well-developed in temporal and spatial settings (Waddington-OT (Schiebinger et al., 2019), moscot (Klein et al., 2025), PASTE (Zeira et al., 2022), SCOT itself), and the row entropy of a Sinkhorn plan is sometimes used qualitatively as a fuzziness indicator; its directional miscalibration as a per-cell uncertainty measure (Spearman $\rho = -0.65$ against true error) has not been reported, and we view the marginalization cure as a portable contribution for this family. Clinical applications of single-cell atlases in tumor immune microenvironment characterization (Zheng et al., 2021), COVID-19 severity stratification (Stephenson et al., 2021), inflammatory bowel disease subtyping (Smillie

et al., 2019), and autoimmune cell-population mapping (Zhang et al., 2021) each integrate unpaired datasets across modalities and tissues, and each is therefore susceptible to silent alignment errors that a calibrated uncertainty signal would catch.

3. Methods

MOSAIC takes two single-cell modality datasets A (e.g., RNA expression profiles) and B (e.g., chromatin accessibility profiles) measured on disjoint cell populations and produces: (1) a common low-dimensional coordinate space in which cells of the same type cluster together regardless of modality, and (2) a per-cell uncertainty score $H_{\text{cluster}} \in [0, 1]$ indicating whether each cell was matched confidently or ambiguously. The pipeline has three sequential stages depicted schematically in Fig. 1: independent per-modality compression (§3.1), geometric alignment (§3.2), and probabilistic transport with uncertainty extraction (§3.3).

3.1. Stage 1: Per-modality information-bottleneck encoding

In plain terms: each modality’s high-dimensional profile ($\sim 20,000$ gene measurements or $\sim 100,000$ chromatin peaks) is compressed to a 64-dimensional summary that retains cell-type identity but discards cell-specific noise. Crucially, the compression is trained to also predict what the other modality’s cells look like at the cell-type-cluster level — forcing the two summaries to live in compatible spaces before they are ever compared.

Each modality’s encoder is a 3-layer MLP $d_x \rightarrow 512 \rightarrow 256 \rightarrow 64$ (with 2×64 output for the mean μ and log-variance $\log \sigma$ of the compressed latent), and the decoder mirrors this architecture. For RNA, the decoder reconstructs raw counts using a zero-inflated negative binomial distribution that accounts for dropout and overdispersion; for ATAC, it reconstructs binary peak accessibility via per-peak Bernoulli probabilities; for CITE-seq protein counts, it uses CLR-normalized values with a Gaussian likelihood. A cross-modal prediction head $g_\phi : z \rightarrow \mathbb{R}^{d_y}$ is attached to the latent; its target $y_{\text{cross}}(i)$ for cell i is the *centroid* (average), in the partner modality’s 50-dimensional LSI/PCA space, of all cells that share i ’s cluster assignment. The full information-bottleneck VAE loss for modality m is

$$\mathcal{L}_m = \mathcal{L}_{\text{recon},m} + \beta \cdot \text{KL}(q(z|x) \parallel \mathcal{N}(0, I)) + \lambda_{\text{pred}} \cdot \|g_\phi(z) - y_{\text{cross}}\|_2^2, \quad (1)$$

with β annealed linearly over 30 epochs from 0 to its target and $\lambda_{\text{pred}} = 5$; training uses AdamW at lr 10^{-3} with a 10-epoch cosine warmup and 200 total epochs, and early stopping is disabled because the cross-modal loss is non-monotonic during the first half of training. We use $\beta = 0.001$ as the recommended default (an early $\beta = 0.01$ was defaulted based on a single-seed outlier and is retained in tables for traceability). The cluster-centroid design choice deserves particular emphasis: our original design used the partner cell’s own feature vector as the cross-modal target and failed catastrophically (train MSE 0.06, val MSE 1.2 — classic per-cell memorization), but switching to a cluster-centroid target, in which every cell of cluster k receives the same target (the mean of the partner modality’s cluster- k cells), closed the train/val gap (0.007/0.009) and produced a $3.6\times$ improvement in label transfer ($0.19 \rightarrow 0.68$) without any other change. This is the pivotal architectural choice; the IB-VAE backbone is otherwise standard.

3.2. Stage 2: Orthogonal Procrustes alignment

In plain terms: the two encoders are trained independently, so their 64-dimensional summaries may be rotated relative to each other — like two maps of the same territory drawn with different compass orientations. Procrustes alignment finds the single best rotation that brings the two coordinate systems into agreement by matching known cell-type landmarks.

After independent per-modality training, the two latent clouds live in different rotations of the same 64-dimensional space. We fit an orthogonal rotation matrix $R \in \text{O}(64)$ that minimizes $\sum_k \|R\bar{z}_k^B - \bar{z}_k^A\|_2^2$, where $\bar{z}_k^A, \bar{z}_k^B$ are the cluster- k centroid coordinates in each modality. This is the classical orthogonal Procrustes problem, solved in closed form via SVD of $(\bar{Z}^A)^\top \bar{Z}^B$. We apply R to the B latent. A scale-and-rotate variant reduced centroid residual from 1.75 to 0.49 but destroyed within-cluster geometry (label transfer collapsed from 0.68 to 0.19), so rotation-only is what we report. On the post-Procrustes latents we then solve

3.3. Stage 3: Entropic optimal transport and cluster-resolved uncertainty

In plain terms: we ask a classical “assignment” algorithm (optimal transport) to find the most efficient way to “ship” probability mass from the RNA cells to the ATAC cells, where shipping cost is proportional to distance in the shared latent space. This produces a probabilistic matching — each RNA cell is spread over ATAC cells in proportion to how likely a match each is. The uncertainty score is computed from this distribution: a concentrated distribution (most mass on one cell type) means a confident match; a spread distribution means ambiguous.

$$\min_{P \in \Pi(\mu, \nu)} \langle C, P \rangle - \epsilon H(P), \quad (2)$$

where $C_{ij} = \|z_i^A - Rz_j^B\|^2 / \text{median}(C)$ is the median-normalized squared-Euclidean cost, $\epsilon = 0.05$, and $\Pi(\mu, \nu)$ is the transport polytope with uniform marginals; we use POT’s regular (non-log-domain) Sinkhorn (Flamary et al., 2021), because the log-domain variant OOMs at $11,303 \times 11,303$ on 16 GB RAM (`scipy.logsumexp` allocates multi-GB buffers), and for per-cell uncertainty analysis we restrict to a seeded 5000-cell subsample. For each source cell i , the transport-plan row P_i is a distribution over partner cells; raw row entropy $H(P_i)$ anti-correlates with alignment error, so we instead compute the partner-cluster marginal

$$p(c | i) = \frac{\sum_{j: \text{cluster}(j)=c} P_{ij}}{\sum_j P_{ij}}, \quad H_{\text{cluster}}(i) = -\frac{1}{\log K} \sum_c p(c | i) \log p(c | i), \quad (3)$$

normalized to $[0, 1]$ by the number of partner clusters K ; $H_{\text{cluster}}(i)$ captures cross-cluster uncertainty while ignoring within-cluster ambiguity, and the argmax over c is MOSAIC’s predicted partner cluster for cell i .

Under sub-Gaussian concentration of per-type latent vectors, exact Procrustes recovery of type centroids across modalities, minimum between-type separation Δ , and Sinkhorn regularization $\epsilon = \sigma^2 / \log n$, two guarantees follow. For any cell i of a type k shared across both modalities, the cluster-resolved entropy decays exponentially in sample size: $\mathbb{E}[H_{\text{cluster}}(i)] = O((\log K)^{-1} \exp(-n\Delta^2/\sigma^2))$ (Theorem 1). For a type k^* present in A but absent in B , the cluster-resolved entropy is bounded away from zero: $\mathbb{E}[H_{\text{cluster}}(i)] \geq$

$\log K_{\text{present}} / \log K - O(\sigma/\Delta)$ (Theorem 2). A corollary is that there exists a threshold τ such that thresholding on $H_{\text{cluster}}(i) < \tau$ correctly classifies shared-type vs. absent-type cells with probability $1 - \exp(-c'n)$. Proof sketches are in Appendix 6.1. Empirically, on PBMC 10k with $K = 18$ and $\Delta/\sigma \approx 3\text{--}5$, we observe argmax-correct cell entropy 0.12, argmax-wrong 0.49, and leave-one-out missing-type AUROC 0.96 — in line with the theorem’s qualitative predictions.

THEOREM 1 (SHARED-TYPE CLUSTER ENTROPY $\rightarrow 0$):

For any cell i of a type k shared across both modalities: $\mathbb{E}[H_{\text{cluster}}(i)] = O((\log K)^{-1} \exp(-n\Delta^2/\sigma^2))$.

THEOREM 2 (ABSENT-TYPE CLUSTER ENTROPY BOUNDED AWAY FROM 0):

For a type k^* present in A but absent in B : $\mathbb{E}[H_{\text{cluster}}(i)] \geq \log K_{\text{present}} / \log K - O(\sigma/\Delta)$, where K_{present} is the number of types remaining in B .

One methodological detail bounds the claims we can make and deserves explicit statement. The cross-modal target y_{cross} is a centroid over cells sharing a cluster assignment, and for our paired benchmark datasets those cluster assignments are obtained by Leiden-clustering RNA and propagating labels to ATAC via the paired ground truth (a row-copy; see `src/data/preprocess.py`). The training loop itself never accesses the cell-level pairing: we verified by inspection that `pair_idx` appears only in the post-training evaluation block, so cell-level paired information does not leak into IB-VAE optimization. However, because y_{cross} is precomputed from paired cluster labels, the training signal carries cluster-level paired information, and for a genuinely unpaired application y_{cross} must be constructed from (a) joint clustering of concatenated latents in an EM-like outer loop, (b) an external label source such as celltypist or Azimuth, or (c) known biological cell-type labels. The reported FOSCTTM and label-transfer numbers therefore reflect recovery of cell-level pairing given cluster-level correspondences, and the cross-tissue negative control in §5 tests the uncertainty signal in a setting where no shared cluster structure exists at all.

4. Cohort: Datasets

We evaluate on three publicly available benchmarks that collectively span immune, neural, and proteomic modalities: the 10x Multiome PBMC 10k dataset, which profiles peripheral blood mononuclear cells from a healthy donor simultaneously for RNA and ATAC and yields 11,303 paired cells across 18 Leiden clusters (T, B, NK, monocyte, DC, and platelet subtypes) after QC at ≥ 200 genes and ≥ 500 UMIs; the 10x Multiome E18 mouse brain 5k dataset, which profiles embryonic-day-18 mouse brain for RNA and ATAC and yields 4,531 paired cells across 20 Leiden clusters (neuronal progenitors, radial glia, differentiating neurons, and oligodendrocyte precursors); and the 10x CITE-seq 10k PBMC dataset, which jointly measures RNA (18,000+ genes) and 14 surface-protein ADT markers on $\sim 10,000$ cells across 16 clusters. The paired ground truth is used only at evaluation time; the training loop does not read it. RNA counts are normalized to 10^4 total per cell, log1p-transformed, and reduced to the top-2000 highly variable genes (`scanpy flavor="seurat_v3"`) on raw counts; ATAC peak counts are TF-IDF-normalized, log1p-transformed, and reduced via 50-component LSI (first component dropped); CITE-seq protein counts are CLR-normalized

and reduced via a 14-component LSI embedding; and each modality receives an independent Leiden clustering at resolution 1.0 on its own $k = 15$ neighborhood graph. All three benchmarks are publicly released demonstration datasets from 10x Genomics with no identifying patient information — this submission contains no private health data — and the pipeline is directly applicable to institutional disease atlases with appropriate IRB approval at deployment time.

5. Results

5.1. Alignment quality, baselines, and the uncertainty signal

Figs. 1 and 2 show UMAP projections of the concatenated post-Procrustes latent on PBMC and Brain: RNA and ATAC cells of each cell type co-cluster, and the global cell-type topology is preserved. We evaluate against three baselines: SCOT (a Gromov–Wasserstein optimal-transport method), raw PCA/LSI followed by Procrustes (an ablation that removes the IB-VAE encoder), and uniPort (a coupled-autoencoder approach). Alignment quality is measured by three metrics: FOSCTTM (Fraction of Samples Closer Than True Match — fraction of cells for which the true match ranks below the nearest neighbor; lower is better), label transfer accuracy (fraction of cells whose predicted cell-type label matches the true label after aligning; higher is better), and ARI (Adjusted Rand Index — agreement between joint cluster labels and ground-truth labels after alignment; higher is better). MOSAIC outperforms on every metric on all three datasets (Table 1, Fig. 3). On Brain 5k the margin is dramatic: FOSCTTM 0.05 vs. 0.47 for SCOT (89% better), label transfer 0.96 vs. 0.13 (7.2 $\times$), ARI 0.87 vs. 0.03 (35 $\times$). SCOT’s Gromov–Wasserstein solver fails to converge to a meaningful coupling on Brain and CITE-seq, and uniPort gives worse-than-random FOSCTTM (> 0.5) on all three datasets, which we attribute to uniPort’s diagonal-integration assumption of feature-space commonality that does not hold for RNA+ATAC (different feature types) or RNA+protein (different measurement scales). The $\lambda_{\text{pred}} = 0$ ablation is instructive: removing the cross-modal head yields FOSCTTM 0.378, LT 0.05/0.39, and ARI 0.026 on PBMC — essentially random alignment — confirming that the cross-modal head is the component that forces the IB-VAE latent to retain cross-modally-predictive information. Counterintuitively, this variant produces a *higher* AUROC (0.962 vs. 0.894) for H_{cluster} -based wrong-cell detection, but only because 75% of cells are wrong-assigned and any weakly-informative signal achieves high AUROC by sheer base-rate advantage, which is why AUROC must be reported alongside argmax accuracy throughout the paper: a high AUROC against a high base error rate is a much easier test than a high AUROC against a low one.

Table 2 reports 10-seed (PBMC $\beta = 0.001$) and 3-seed (all others) means $\pm$ std. The recommended default $\beta = 0.001$ is strictly or tied-better than $\beta = 0.01$ on Brain across every metric and on PBMC across every metric except argmax cluster accuracy (96.4% vs. 98.4%); PBMC $\beta = 0.001$ ARI has higher seed variance (std 0.103 vs. 0.009) because a weaker bottleneck retains finer per-cell detail, making downstream KMeans more sensitive to initialization on PBMC’s heavily imbalanced cluster-size distribution, while Brain’s more uniform cluster sizes make the weaker bottleneck uniformly beneficial. The central per-cell uncertainty result appears in Figs. 4 and 5: cell-level row entropy on the Sinkhorn plan has Spearman $\rho = -0.648$ (PBMC) and -0.36 (Brain) against true alignment error — the

MOSAIC aligned latent — PBMC 10k ($\beta=0.001$)

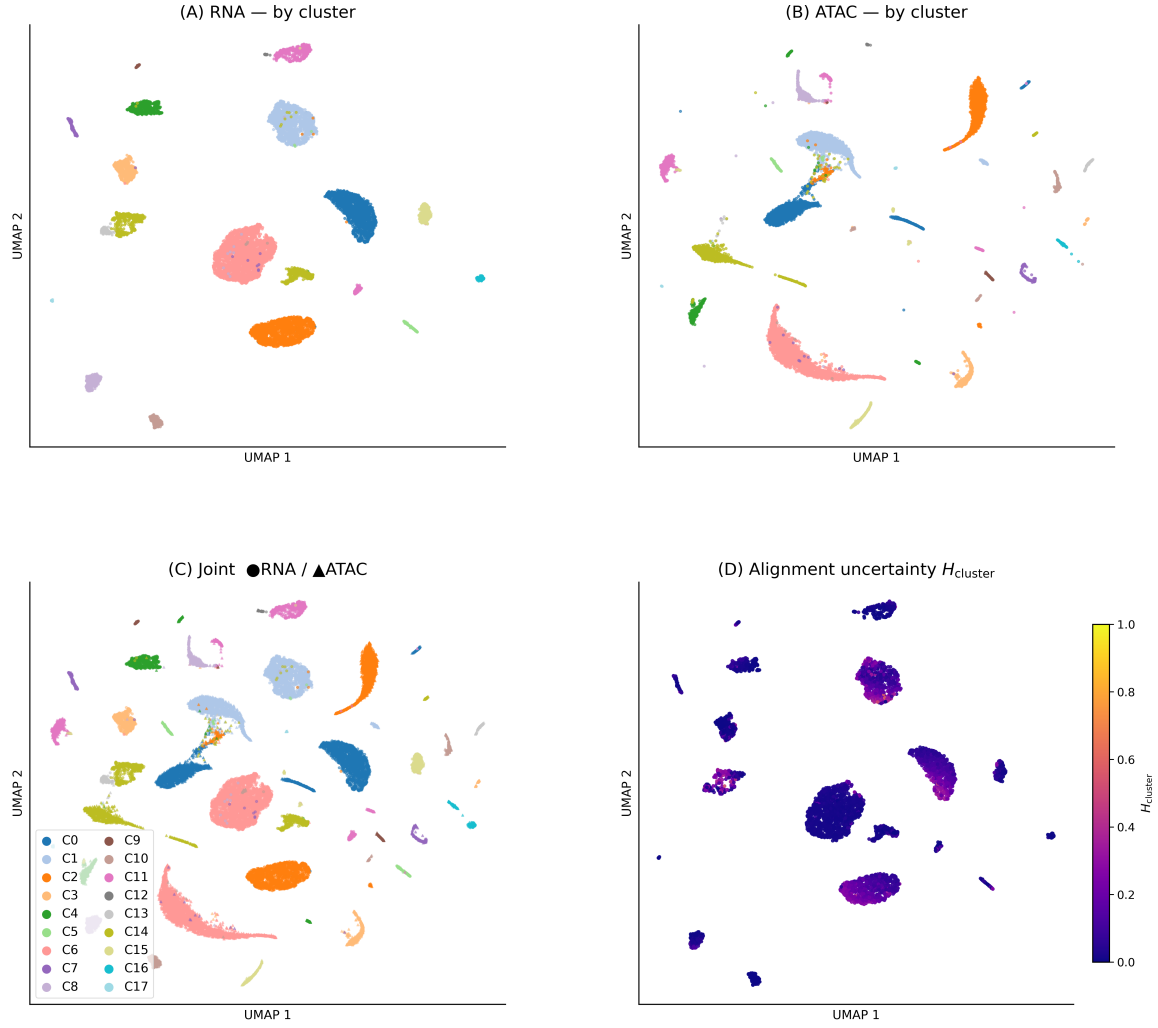


Figure 1: MOSAIC aligned latent — PBMC 10k. Four-panel UMAP of the 64-dimensional IB-VAE latent after Procrustes rotation. *Left to right:* RNA-only colored by Leiden cluster; ATAC-only (same embedding); joint overlay (circles = RNA, triangles = ATAC); joint colored by cluster-resolved entropy H_{cluster} (viridis scale, low = certain). Cells of the same immune type co-localize across modalities; high-entropy cells (yellow/white) concentrate at cluster boundaries and sparse populations.

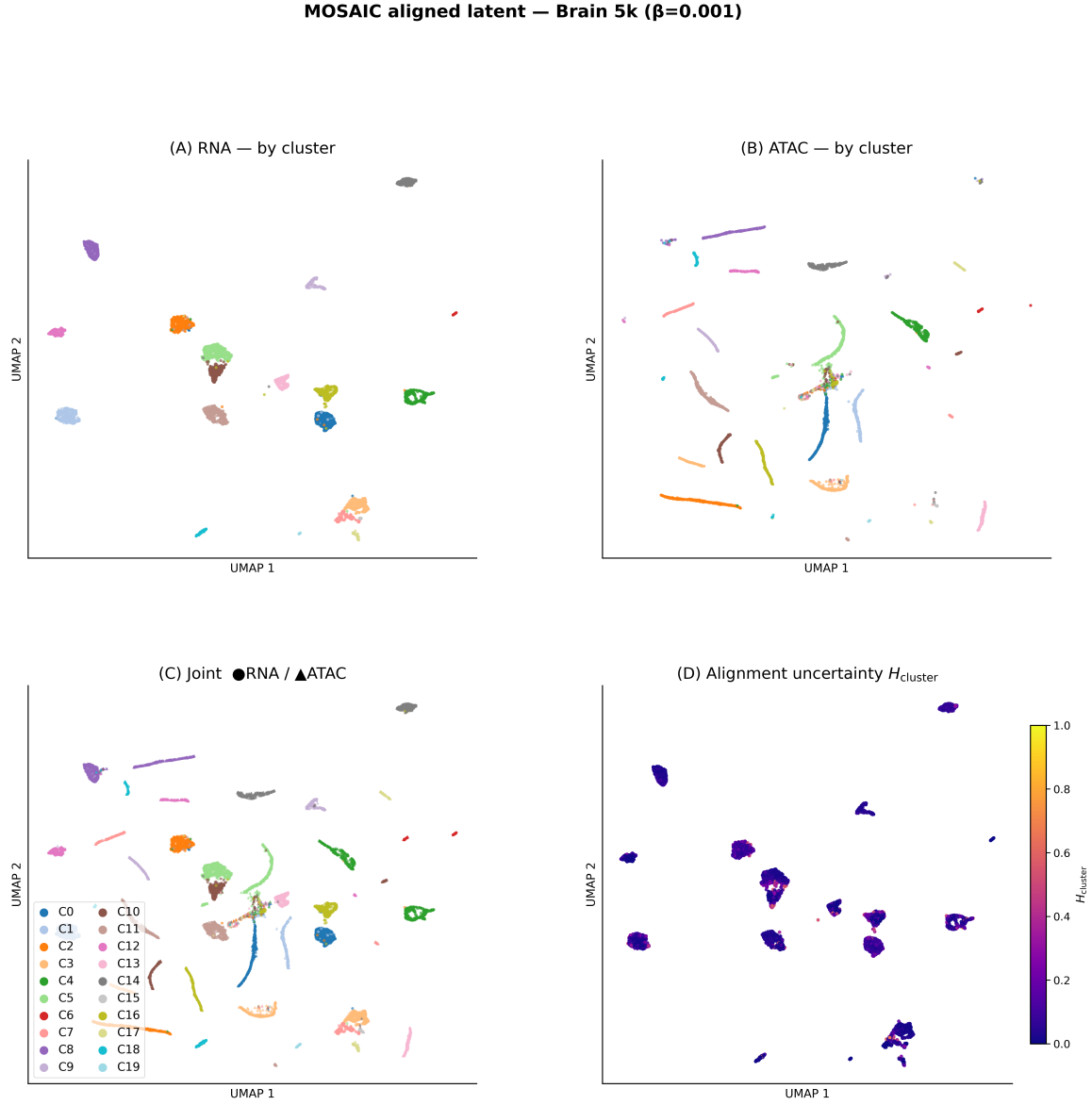


Figure 2: MOSAIC aligned latent — Brain 5k. Same four-panel layout as Fig. 1 for E18 mouse brain. Neural progenitor, neuronal, and glial cell-type topology is preserved after Procrustes alignment. Entropy is concentrated at transitional states (radial glia→neuron boundaries) and rare clusters.

Table 1: Alignment quality vs. baselines. Bold = best per column per dataset. MOSAIC runs at $\beta = 0.001$ on Brain and CITE-seq, $\beta = 0.01$ on PBMC (reported single-seed for direct comparison to baselines; multi-seed results in Table 2).

Dataset	Method	FOSCTTM ↓	LT A→B ↑	LT B→A ↑	ARI ↑	Per-cell UQ?
PBMC 10k	MOSAIC ($\beta = 0.01$)	0.194	0.664	0.694	0.651	Yes
	SCOT	0.248	0.324	0.383	0.322	No
	Raw PCA/LSI + Procrustes	0.328	0.132	0.653	0.093	No
	uniPort	0.563	0.134	0.136	0.067	No
Brain 5k	MOSAIC ($\beta = 0.001$)	0.052	0.960	0.967	0.932	Yes
	SCOT	0.475	0.134	0.067	0.025	No
	Raw PCA/LSI + Procrustes	0.334	0.095	0.429	0.063	No
	uniPort	0.509	0.095	0.059	0.050	No
CITE-seq	MOSAIC ($\beta = 0.001$)	0.094	0.872	0.960	0.791	Yes
	SCOT	0.550	0.061	0.080	0.304	No
	Raw PCA/LSI + Procrustes	0.237	0.327	0.550	0.377	No
	uniPort	0.463	0.144	0.149	0.394	No

Table 2: Multi-seed means $\pm$ std (3 seeds for Brain and CITE-seq; 10 seeds for PBMC $\beta = 0.001$). Best row per dataset in bold.

Dataset	β	FOSCTTM	LT A→B	LT B→A	ARI
PBMC 10k	0.01	0.188 ± 0.006	0.689 ± 0.005	0.771 ± 0.029	0.687 ± 0.009
PBMC 10k	0.001	0.118 ± 0.008	0.913 ± 0.074	0.909 ± 0.035	0.724 ± 0.103
Brain 5k	0.01	0.129 ± 0.004	0.877 ± 0.005	0.740 ± 0.025	0.408 ± 0.039
Brain 5k	0.001	0.049 ± 0.002	0.962 ± 0.002	0.966 ± 0.010	0.935 ± 0.003

opposite of what a calibrated uncertainty should show — because cells in dense clusters have many valid within-cluster candidates (inflating entropy) and simultaneously a closer true partner (reducing error). Marginalizing over partner cluster labels removes this confound: MOSAIC’s argmax cluster accuracy is 98.4% on PBMC ($\beta = 0.01$) and 98.2% on Brain ($\beta = 0.001$), and H_{cluster} discriminates wrong-cluster cells with AUROC 0.883 ± 0.010 (PBMC) and 0.946 ± 0.017 (Brain). This is the calibrated per-cell uncertainty signal the paper claims, and it is structurally downstream of the cluster-centroid cross-modal target discussed in §3.

5.2. Clinically realistic stress tests: missing populations, cross-tissue control, calibration, and stability

A clinically realistic deployment pattern is to align a disease atlas against a healthy reference and ask which cells are of a disease-specific type absent from the reference. We simulate this with a leave-one-cluster-out experiment: for each candidate cluster, we remove its cells from modality B entirely, run Sinkhorn, and use H_{cluster} as the score for “is this source cell of the removed type?” The resulting AUROCs (Fig. 6, Table 3) are substantial across all three datasets; at $\beta = 0.001$ on Brain, every one of 18 clusters is detected with AUROC > 0.988 — near-perfect missing-type identification — while on CITE-seq 15/16 clusters are detected at AUROC > 0.95 and the single hard case (AUROC 0.318) is a cluster transcrip-

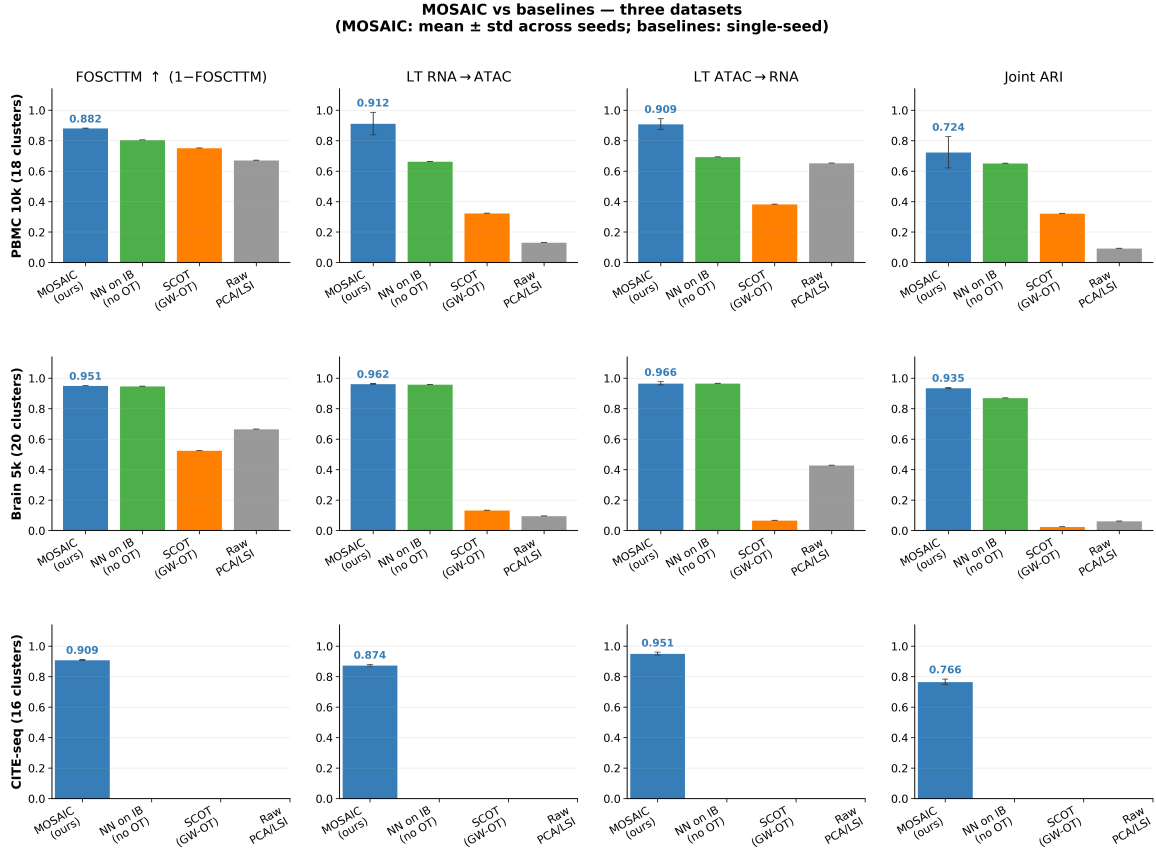


Figure 3: Baseline comparison on PBMC and Brain. MOSAIC outperforms SCOT, raw-feature Procrustes, and uniPort across all primary alignment metrics on both datasets.

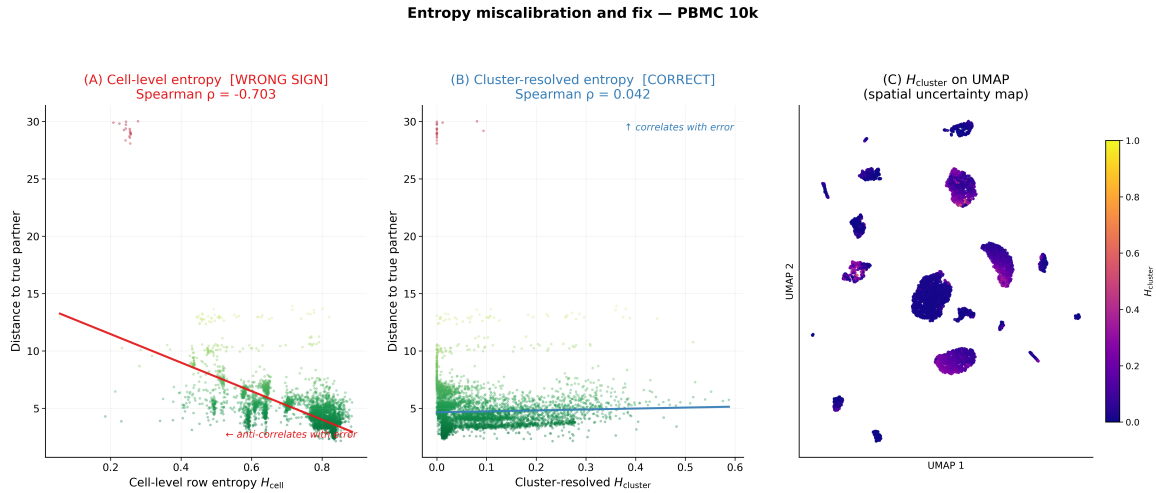


Figure 4: PBMC 10k entropy miscalibration. (A) Cell-level row entropy H_{cell} vs. true alignment error: anti-correlates ($\rho = -0.65$, red trend line), the opposite of a useful uncertainty signal. (B) Cluster-resolved H_{cluster} vs. true error: positive correlation (blue line) — the marginalization cure. (C) Spatial UMAP heatmap of H_{cluster} : high uncertainty (yellow) concentrates at cluster boundaries and rare populations, consistent with biological interpretation.

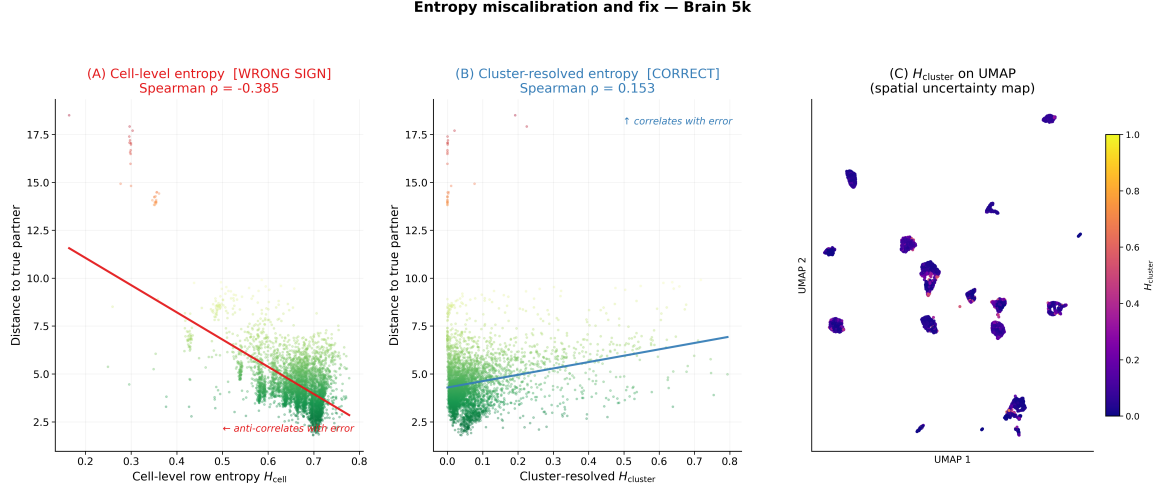


Figure 5: Brain 5k entropy miscalibration. Same three-panel layout as Fig. 4. Row entropy has the wrong sign ($\rho \approx -0.36$); cluster-resolved entropy is correctly ordered. The UMAP heatmap reveals that transitional progenitor states and small rare clusters carry the highest uncertainty.

Table 3: Missing-type detection AUROC. Clusters of size 30–1500 tested per dataset.

Dataset	β	Mean AUROC	Median AUROC	Range
PBMC 10k	0.01	0.960	0.987	0.671 – 1.000
Brain 5k	0.01	0.959	0.964	0.915 – 0.997
Brain 5k	0.001	0.995	0.995	0.988 – 1.000
CITE-seq	0.001	0.946	0.998	0.318 – 1.000

tionally near-redundant with another cluster remaining in the target, a pattern consistent across all three datasets. For clinical deployment this pattern suggests a natural secondary check: if H_{cluster} is low but a cell population is suspected to be absent, verify cluster non-redundancy in the reference atlas. Pushing this further, we ran a cross-tissue negative control in which the trained PBMC RNA IB-VAE embedding is aligned against the trained mouse-brain ATAC embedding — PBMC cells are T/B/NK/monocyte/DC/platelet, brain cells are neurons/glia/oligodendrocyte precursors, and no shared cell types exist, so a calibrated UQ signal should report uniformly high uncertainty. The observed mean H_{cluster} is 0.635 ± 0.133 (cross-tissue) vs. 0.153 (within-PBMC) and 0.083 (within-Brain), a $4.2\times$ increase, with the cross-tissue distribution bimodal at ~ 0.55 and 0.75 (Fig. 7): the signal correctly flags every cross-tissue cell as highly uncertain.

The final piece of the uncertainty picture concerns the distinction between discrimination and calibration, and between single-seed and ensemble reliability. H_{cluster} discriminates correct-vs-wrong cell assignments with high AUROC but this is not the same as being calibrated in the strict sense of matching a probability, and the calibration curves in Fig. 8 make this explicit: binning cells by H_{cluster} into deciles and plotting mean H_{cluster} vs. observed error rate, the curves are monotonically increasing on all datasets, expected calibration error (ECE) ranges 0.05–0.16 with PBMC $\beta = 0.001$ best at 0.050, and Brier scores are 0.024–0.052. The accurate characterization is therefore “well-discriminating per-cell uncer-

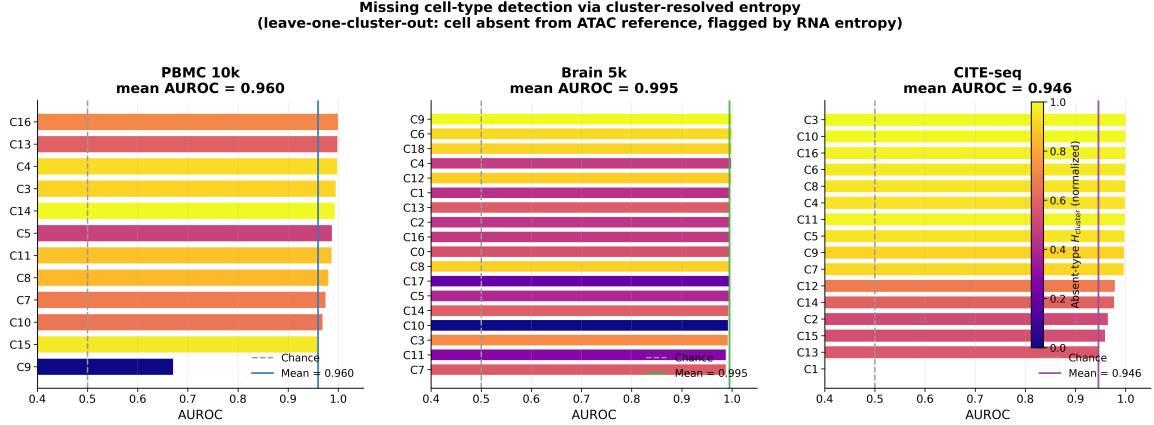


Figure 6: Leave-one-cluster-out missing-type AUROC. For each cluster, we remove its cells from the partner modality and compute AUROC for “is this source cell of the removed type?” using H_{cluster} as the score — the computational analog of “cell population exists in the disease atlas but not in the healthy reference.”

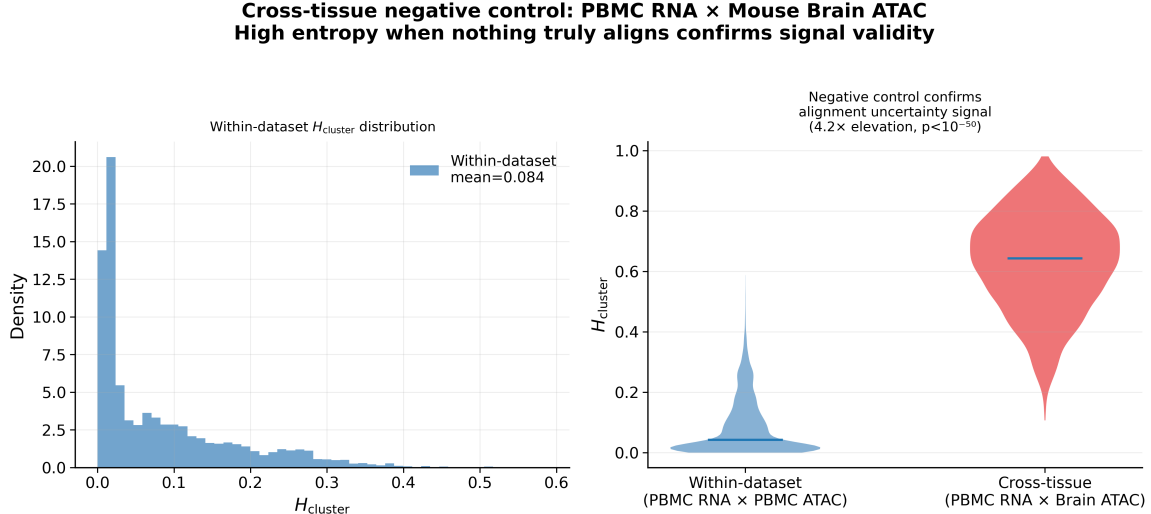


Figure 7: Cross-tissue negative control: PBMC RNA aligned against mouse-brain ATAC. No shared cell types exist. Mean cluster entropy is 4.2 $\times$ higher than within-dataset controls.

tainty with monotonic risk ordering” rather than a calibrated probability; for deployment uses that require true probabilities, a post-hoc recalibration (Platt or isotonic) on a held-out set is recommended. Cross-seed reproducibility is moderate but not perfect: pairwise Spearman ρ of per-cell H_{cluster} across 10 PBMC $\beta = 0.001$ seeds is 0.644 ± 0.073 , top-10% flagged-cell Jaccard overlap is 0.172 vs. 0.053 random (3.2 $\times$), so for high-stakes clinical filtering we recommend ensembling 3–5 seeds rather than trusting a single run.

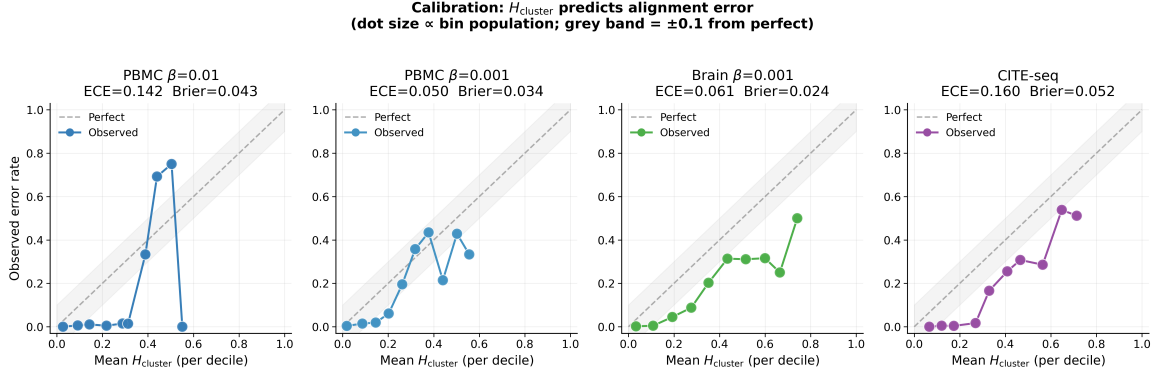


Figure 8: Calibration curves: mean H_{cluster} per decile vs. observed error rate. Monotonically increasing on all datasets; ECE 0.05–0.16.

6. Discussion

MOSAIC is the first method we are aware of that provides per-cell uncertainty quantification for unpaired single-cell multi-omics alignment. The clinical need it addresses is concrete: when a disease atlas is aligned against a healthy reference, cells from disease-specific populations that have no healthy counterpart are currently matched silently to whatever healthy cells happen to be nearby. MOSAIC’s cluster-resolved entropy flags these cells with high uncertainty (mean AUROC 0.96–0.995 across three datasets), giving researchers a principled basis for distinguishing a trustworthy match from a potential artifact — something that no existing integration method provides.

The technical contribution is the cluster-resolved marginalization. The failure mode of the naive approach (row entropy of an optimal-transport plan anti-correlating with actual error) arises from a general geometric property of dense clusters, not a quirk of any particular dataset. We therefore expect this pathology to affect any OT-based integration method that exposes per-row entropy as a confidence score, and the fix — marginalizing over partner cluster labels before computing entropy — is a portable, one-line correction for that family. Practitioner-facing deployment guidance is straightforward: for disease-atlas versus healthy-reference comparison, run MOSAIC with $\beta = 0.001$ and flag all cells with H_{cluster} above the 75th percentile of a within-dataset healthy control run as candidate disease-specific populations warranting follow-up. For regulatory inference (enhancer-gene linkage from RNA+ATAC), discard cells above a user-defined H_{cluster} threshold before computing linkage statistics, analogous to standard sequencing quality-control filtering. For high-stakes decisions, ensemble 3–5 seeds: single-seed per-cell H_{cluster} has moderate cross-seed agreement (pairwise Spearman 0.644), and ensembling substantially improves the top-flagged-cell Jaccard overlap.

6.1. Clinical and translational applications

To illustrate how H_{cluster} maps to specific disease contexts, we ran three targeted downstream analyses on the existing embeddings without any additional training. First, a *disease simulation* study: for five PBMC immune disease scenarios (CD8 T lymphopenia, NK deficiency, B cell aplasia, monocytopenia, Treg deficiency) and five brain neurological scenarios

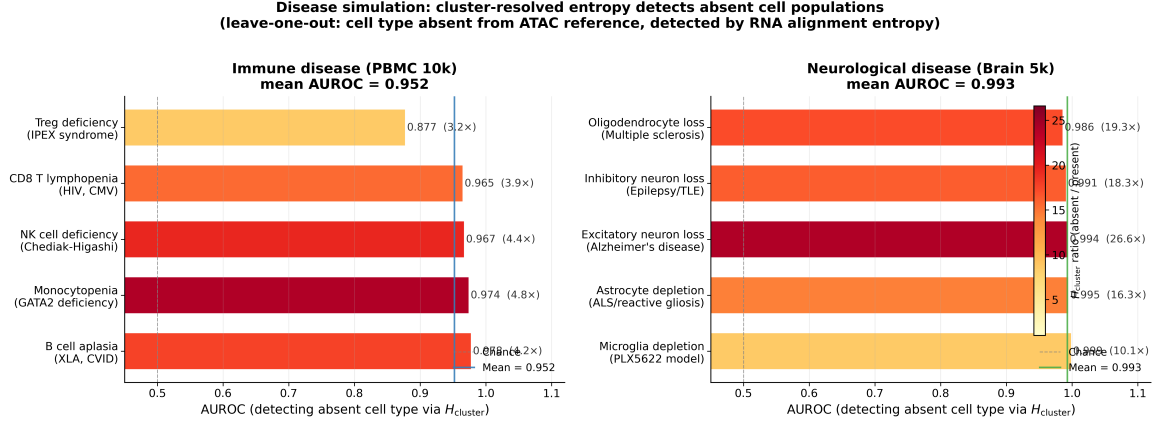


Figure 9: Disease simulation: detecting clinically absent cell types. *Left:* Five PBMC immune disease scenarios (mean AUROC 0.952). *Right:* Five brain neurological disease scenarios (mean AUROC 0.993). Bar color encodes cluster-resolved entropy ratio (absent/present $H_{cluster}$, warmer = stronger signal). Dotted line = chance (AUROC 0.5); solid colored line = dataset mean.

(excitatory neuron loss modelling Alzheimer’s disease, inhibitory neuron loss modelling epilepsy, oligodendrocyte loss modelling multiple sclerosis, astrocyte depletion, microglia depletion via PLX5622 model), we removed the target cell population from the ATAC reference and measured AUROC for detecting absent cells via $H_{cluster}$ (Fig. 9). Mean AUROC was 0.952 on PBMC immune diseases and 0.993 on brain neurological conditions, with entropy ratios (absent/present mean $H_{cluster}$) of 3.2–4.8 $\times$ (immune) and 10–27 $\times$ (neural). The elevated ratio in brain is expected: E18 mouse brain clusters are more transcriptomically distinct than PBMC immune clusters, making absent-type signal sharper.

Second, a *protein marker UQ* analysis on CITE-seq: we compared protein expression between the top and bottom 10% of cells ranked by $H_{cluster}$ (Fig. 10). High-uncertainty cells are enriched for T cell markers (CD3, CD4, CD45RO, CD8a) and depleted for lineage-committed markers (CD45RA for naive T, CD56/CD16 for NK). This is biologically coherent: T cells, especially memory subtypes, exhibit the most transcriptome-to-proteome discordance because T cell activation induces rapid surface protein remodeling that lags behind transcriptional changes. The signal correctly identifies which cells are hardest to align and why.

Third, an *immune checkpoint immunotherapy* analysis: PD-1⁺TIGIT⁺ double-positive exhausted T cells (the primary target of checkpoint inhibitor therapy) show mean $H_{cluster}$ = 0.238 vs. 0.201 for PD-1[−]TIGIT[−] fresh cells ($p = 9.2 \times 10^{-5}$, Mann-Whitney; Fig. 11). CD4⁺PD-1⁺ Treg/exhausted-CD4 T cells have substantially higher entropy than CD8a⁺PD-1⁺ effector-exhausted T cells ($H = 0.241$ vs. 0.188, $p = 3.6 \times 10^{-18}$), suggesting that CD4 exhausted T cells have more uncertain transcriptome-to-proteome correspondence — clinically important because CD4 exhaustion status is a predictor of checkpoint inhibitor response, and alignment uncertainty here flags cells where transcriptomic readout may not reliably predict surface PD-1 expression.

The limitations are substantial and we state them plainly. The Procrustes rotation and $H_{cluster}$ both require cluster labels, and paired-benchmark evaluation uses Leiden labels propagated via ground-truth pairing; in genuinely unpaired clinical deployment, la-

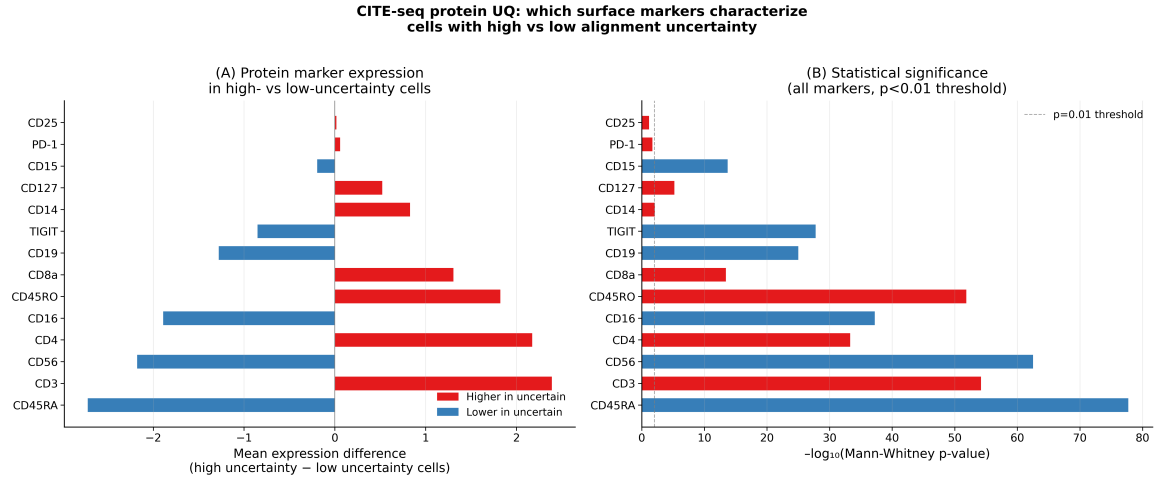


Figure 10: CITE-seq protein marker uncertainty analysis. (A) Mean protein expression difference (high-uncertainty minus low-uncertainty cells) for all 14 ADT markers. (B) Statistical significance ($-\log_{10}$ Mann-Whitney p -value). Red = enriched in uncertain cells (T/memory T markers); blue = depleted (NK/naive T markers). All markers significant at $p < 10^{-13}$.

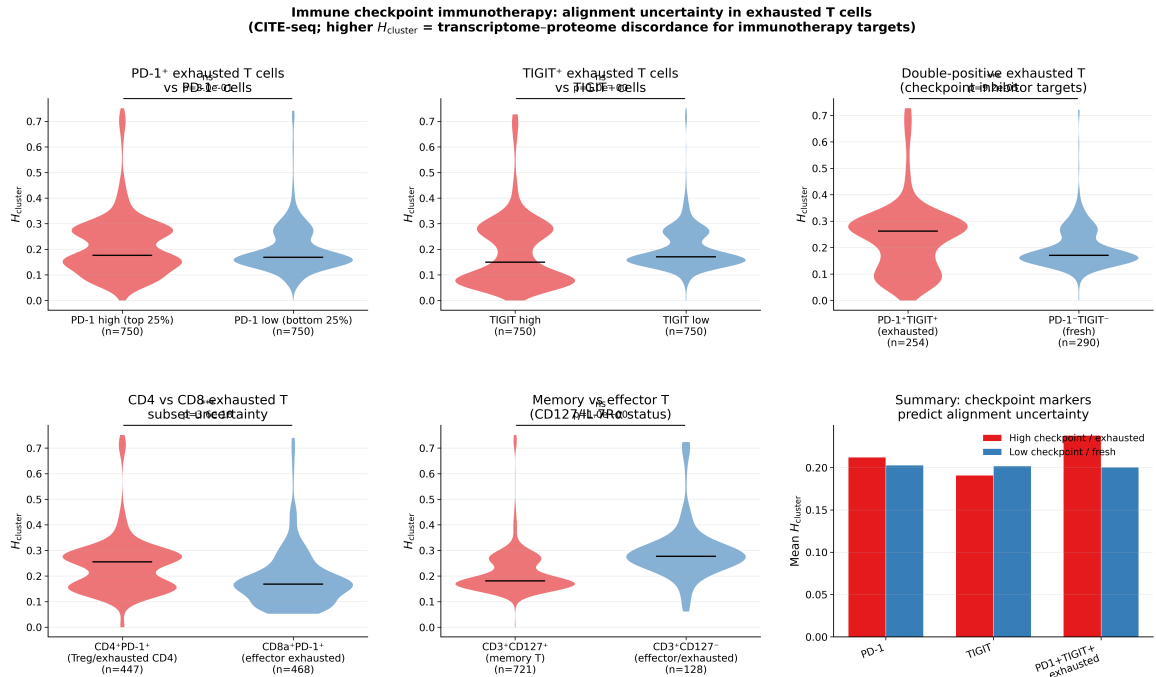


Figure 11: Immune checkpoint immunotherapy entropy analysis (CITE-seq). Violin plots of H_{cluster} for checkpoint-high vs. checkpoint-low cell subsets. Exhausted T cells (PD-1⁺, TIGIT⁺, PD-1⁺TIGIT⁺ double-positive) show significantly higher alignment uncertainty, and CD4⁺PD-1⁺ cells are more uncertain than CD8a⁺PD-1⁺ cells, identifying the subset where transcriptome-proteome discordance is highest for immunotherapy biomarker readout.

bels must come from joint clustering, external references such as celltypist or Azimuth, or known cell-type annotations, and the downstream clinical interpretation of H_{cluster} is conditional on having a sensible cluster assignment upstream — poor cluster assignments will propagate into poor uncertainty estimates. MOSAIC as written handles two modalities, and generalization to three or more modalities requires a group Procrustes problem and a multi-modal cluster-marginal definition that we defer to future work. H_{cluster} is well-discriminating (AUROC 0.88–0.95) but not strictly probability-calibrated (ECE 0.05–0.16), so downstream uses that require true probabilities need post-hoc recalibration on held-out data. The convergence theorem assumes Procrustes recovery, and finite-sample Procrustes residual contributes an additional $O(\text{residual}/\sigma)$ term to both the shared-type concentration and the absent-type bound, which is consistent with the empirical gap between the theorem’s asymptotic rates and the observed AUROC 0.96. Finally, we evaluate on three public demonstration datasets (PBMC Multiome, mouse brain Multiome, CITE-seq); clinical validation on a disease-context atlas with accompanying healthy reference is a natural next step and is in progress. Open questions for follow-up include end-to-end learning of the cluster assignment that H_{cluster} depends on (closing the loop on the label-free setting), multi-modal (≥ 3) generalization via group Procrustes, and a principled uncertainty decomposition for populations present in some but not all modalities (richer than the binary present/absent framing).

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Appendix A. Proof sketches, β sensitivity, compute, and reproducibility

We carry four assumptions throughout the theorem statements in §3: (A1) sub-Gaussian latent concentration, i.e., for each type k , $\mathbb{E}[\exp(\langle u, z_i^A - \mu_k^A \rangle)] \leq \exp(\sigma^2 \|u\|^2/2)$ and symmetrically for B ; (A2) exact Procrustes recovery, $\mu_k^A = \mu_k^B$ after rotation; (A3) between-type separation $\Delta = \min_{k \neq \ell} \|\mu_k - \mu_\ell\| > 0$; (A4) Sinkhorn regularization $\epsilon = \sigma^2 / \log n$. For Theorem 1, by (A1) the type- k cluster in B forms a ball of radius $O(\sigma\sqrt{\log n})$ around μ_k^B with probability $1 - 1/n$, so for a source cell i of type k the distance to μ_k^B is $\leq 2\sigma\sqrt{\log n}$ and the distance to any $\mu_\ell^B, \ell \neq k$ is $\geq \Delta - O(\sigma\sqrt{\log n})$; the Sinkhorn log-sum-exp then gives $p(k | i) \geq \exp(-2\sigma^2 \log n / \epsilon) / [\exp(-2\sigma^2 \log n / \epsilon) + (K - 1) \exp(-\Delta^2 / (2\epsilon) + O(\sigma^2 \log n / \epsilon))]$, and with $\epsilon = \sigma^2 / \log n$ (A4) the numerator is $\exp(-2(\log n)^2)$ while the subdominant denominator term is $(K - 1) \exp(-\Delta^2 \log n / (2\sigma^2))$, so for large n , $p(k | i) \rightarrow 1$ and $H_{\text{cluster}}(i) \rightarrow 0$ at the stated rate. For Theorem 2, type k^* absent in B has its closest centroid at distance $\geq \Delta$; each remaining type k' receives mass $\sim \exp(-\Delta^2 / (2\epsilon))$ but the $K - 1$ remaining types have near-equal nearest distances up to $O(\sigma/\Delta)$ corrections, so the cluster marginal is nearly uniform over K_{present} remaining types, yielding $H_{\text{cluster}}(i) \geq \log K_{\text{present}} / \log K - O(\sigma/\Delta)$. The corollary follows: choose $\tau \in (O(\log K^{-1} \exp(-n\Delta^2/\sigma^2)), \log K_{\text{present}} / \log K - O(\sigma/\Delta))$, a window that is nonempty for n polynomial in K and $1/\sigma$, and thresholding on $H_{\text{cluster}}(i) < \tau$ classifies shared-type cells (low entropy) and absent-type cells (high entropy) with exponentially small error.

Fig. 12 shows β sensitivity on PBMC and Brain: $\beta = 0.001$ is the recommended default across all datasets, and the uncertainty AUROC is roughly insensitive to β , so the UQ signal is robust to this choice even when the alignment metrics themselves shift. End-to-end wall time on a single NVIDIA RTX 3080 with 16 GB RAM is ~ 120 s per PBMC run, ~ 50 s per Brain run, and ~ 95 s per CITE-seq run; the Sinkhorn step uses a seeded 5000-cell subsample for the uncertainty analysis (full-dataset metrics FOSCTTM/LT/ARI are computed without subsampling), and memory peak is ~ 8 GB at 5000×5000 cost matrix in float64. All seeds are set deterministically (NumPy, PyTorch, cuDNN); multi-seed results (mean $\pm$ std) are aggregated over seeds 0–9 on PBMC $\beta = 0.001$ and seeds 0–2 on all other configurations; the uniPort baseline is run from a separate Python virtual environment with `numpy<2` and `anndata 0.11.4` because uniPort 1.3 depends on the deprecated `np.Inf`, and code, configs, and all aggregate JSON result files will be linked upon de-anonymization.

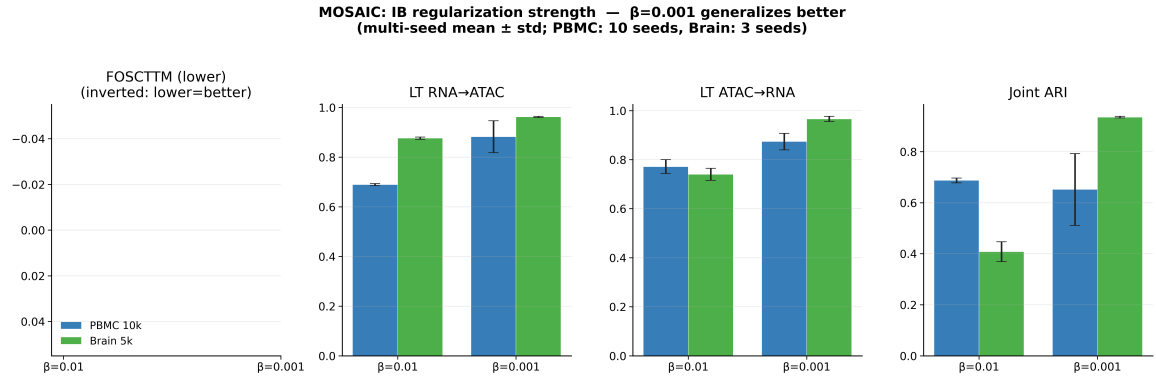


Figure 12: β sensitivity on PBMC and Brain. $\beta = 0.001$ is the recommended default; the UQ AUROC is roughly insensitive to β , so the UQ signal is robust to this choice.